

Yahia Kerroum

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About Me

Fourth-year AI engineering student at ENSIA with hands-on experience building full-stack AI systems, web platforms, mobile applications, and ML pipelines. Focused on practical, deployable solutions across computer vision, recommendation systems, adaptive learning, and ecosystem tooling.

Education

National Higher School of Artificial Intelligence (ENSIA)

Expected 2028

Master & Engineer's Degree in Artificial Intelligence, Algiers, Algeria

Relevant Coursework: Machine Learning, Data Mining, Statistics & Probability, Computer Vision, Natural Language Processing, Web Development, Mobile Development, Data Structures and Algorithms, Database Systems, Artificial Intelligence, Object-Oriented Programming, Linux.

Technical Skills

AI and Machine Learning: Machine Learning, Deep Learning, Neural Networks, Computer Vision, NLP, Statistics, Probability, Federated Learning, Active Learning, Reinforcement Learning, Recommendation Systems, Heuristic Search, Genetic Algorithms, Constraint Satisfaction

Data Science Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Feature Engineering, Data Preprocessing

Software Engineering: Full-Stack Web Development, REST API Design, Mobile Development, Clean Architecture, Design Patterns, Agile Methodology, WebSockets, Web Scraping

Programming Languages: Python, TypeScript, JavaScript, C++, Dart, SQL

Databases and Infrastructure: PostgreSQL, Redis, Firebase, Docker, Git, Linux, Nginx

Projects

DzairAI – Algerian AI Ecosystem Platform

Major Contributor | 2025 – 2026

- Contributed to a large-scale production AI ecosystem platform centralizing Algeria's AI landscape across researchers, organizations, events, datasets, research papers, and open-source repositories, built for self-hosted Algerian deployment across a modular architecture spanning 72 API routes and 419 source files.
- Built automated Python scrapers for Algerian GitHub repositories, academic papers, and AI news, storing structured results in PostgreSQL via SQL pipelines; contributed to multilingual CV parsing and AI-driven job matching pipelines.
- Platform provides unified search, authenticated profiles, bookmarks, collaboration requests, and dataset preview workflows designed to scale to national usage.

Dental Clinic Management System (DCMS)

Lead | 2025

- Led development of a production-ready clinic platform with a dynamic permission system of 50+ granular permissions in resource.action format, allowing managers to modify staff access at runtime without redeployment.
- Built an interactive dental orthogram with graphical tooth mapping (1 to 32 numbering, color-coded indicators), multi-session treatment tracking with clinical notes, and appointment conflict detection across doctors.
- Integrated Google Drive API for patient-linked document storage, 18+ analytics report types with role-personalized dashboards, payment and expense approval workflows, and full CSV export for regulatory compliance.

Findar – Real Estate Mobile Application

Major Contributor | 2025

- Contributed to a full-stack Flutter mobile app for the Algerian real estate market enabling sellers, agencies, buyers, and renters to list and discover properties across houses, apartments, villas, and studios.
- Features Firebase-backed authentication with OAuth, real-time push notifications, listing boosts for visibility promotion, saved favorites, and a dedicated public landing page. Available as an Android APK.

AI Job Matching Platform

Major Contributor | 2025

- Contributed to an AI-driven matching system processing 8,500+ job listings and 8,100+ employee profiles stored in PostgreSQL; personally implemented the genetic algorithm, heuristic function, and A* and CSP problem

formulation in Python.

- Four algorithms benchmarked for bidirectional matching across skills, experience, education, city, salary, and sector: A* at 92% accuracy, Genetic Algorithm at 85%, Greedy at 78%, and CSP for hard-constraint scenarios.
- Used Pandas and NumPy for feature vectorization and SQL queries for candidate-job filtering; hierarchical feature-based clustering reduced search space by 60%, achieving sub-5ms end-to-end latency across all strategies.

Sentinel – Browser-Native Phishing Detection Extension

Contributor | 2026

- Contributed to a privacy-preserving Chrome extension running all phishing analysis in-browser with no email content transmitted externally, combining offline MiniLM ONNX inference, Damerau-Levenshtein typosquatting detection, RDAP domain-age lookups, and inline Gmail badge annotations.

FaceAttend – Secure Face Recognition Attendance System

Major Contributor | 2026

- Contributed to a two-tier anti-spoofing pipeline where client-side MediaPipe Face Mesh running in-browser and a server-side MobileNetV2 CNN must both be defeated simultaneously for an attack to succeed.
- Worked on ArcFace 512-dimensional embedding generation and FAISS cosine similarity search against pgvector for real-time identity verification, with Redis-based recognition cooldowns to prevent double-marking.
- System includes role-based access for admins, teachers, and students, WebSocket live attendance updates, an immutable audit log, and a student dispute workflow.

Adaptive Quiz Platform

Lead | 2025 – 2026

- Led an ML pipeline in Python (Pandas, NumPy, Scikit-learn) to classify student knowledge level using as few questions as possible, covering multiple CS modules with balanced difficulty labeling and near-duplicate filtering.
- Benchmarked active learning strategies (Uncertainty Sampling, Entropy Reduction, Query-by-Committee) against a random baseline, proving adaptive selection converges to accurate classification faster; validated against real ENSIA student data via statistical analysis of accuracy and convergence rates.

Languages

Arabic (Native) French (Professional) English (Professional)